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THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Boundedness of the derived lower shriek functors (Lemma lemma-derived-lower-shriek-bounded, canonical tag 0G2A). cited at proof lines 4114 and 4119. Supplies the integer N and the vanishing of Rf_! of the truncation in degrees <= a - 1.
- Negative exts vanish (Derived Categories, Lemma derived-lemma-negative-exts, canonical tag 06XS). cited at proof lines 4121-4122. Hom from a complex vanishing above degree a - 1 into K vanishing below degree a is zero.

Source reference. Original lemma 0G2A: sources/stacks-chow/tags.txt, source lines 20050–20050. Cited at proof lines 4114 and 4119: supplies the integer N and the vanishing of Rf_! of the truncation in degrees <= a - 1.

Source reference. Original lemma 06XS: sources/stacks-chow/tags.txt, source lines 8861–8861. Cited at proof lines 4121-4122: Hom from a complex with vanishing cohomology above degree a - 1 into K vanishing below degree a is zero.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion ring. The functor $Rf^!$ sends $D^+(Y_{étale}, \Lambda)$ into $D^+(X_{étale}, \Lambda)$. More precisely, there exists an integer $N \geq 0$ such that if $K \in D(Y_{étale}, \Lambda)$ has $H^i(K) = 0$ for $i < a$ then $H^i(Rf^!K) = 0$ for $i < a - N$.*

Proof. Let N be the integer found in Lemma 0G2A. By construction, for $K \in D(Y_{étale}, \Lambda)$ and $L \in\in D(X_{étale}, \Lambda)$ we have $\mathrm{Hom}_X(L, Rf^!K) = \mathrm{Hom}_Y(Rf_!L, K)$. Suppose $H^i(K) = 0$ for $i < a$. Then we take $L = \tau_{\leq a-N-1}Rf^!K$. By Lemma 0G2A the complex $Rf_!L$ has vanishing cohomology sheaves in degrees $\leq a - 1$. Hence $\mathrm{Hom}_Y(Rf_!L, K) = 0$ by Derived Categories, Lemma 06XS. Hence the canonical map $\tau_{\leq a-N-1}Rf^!K \to Rf^!K$ is zero which implies $H^i(Rf^!K) = 0$ for $i \leq a - N - 1$. □

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